

CloudWatch

CloudWatch serves as AWS's primary monitoring service that provides visibility into resources and applications, helping track metrics like website traffic and performance

Key concepts include:

- Namespaces
- Function as containers for metrics
- Include standard options for AWS services (like DynamoDB, EBS)
- Can be customized but are region-specific^s
- Monitoring Levels
- Basic monitoring: Free, 5-minute intervals, automatically enabled
- Detailed monitoring: Paid, 1-minute intervals, requires manual activation^s
- Metric Resolution
- Standard resolution: 1-minute granularity
- High resolution: 1-second accuracy
- Console Features
- Ability to filter and view specific metrics
- Customizable time ranges and visualization options
- Capability to set up alarms based on metric thresholds

Overview of Amazon CloudWatch

- CloudWatch is a core service in AWS that provides visibility into AWS resources and applications, essential for monitoring applications deployed on EC2 instances and databases.
- It helps answer questions about website traffic, performance, bandwidth usage, and optimization of compute resources.

CloudWatch Metrics

- Metrics are stored in namespaces, which are containers for the metrics, with standard namespaces like Dynamo DB and EBS available.
- Custom namespaces can also be created, but metrics are region-specific^s.
- Metrics consist of time-ordered sets of data points that change over time, each accompanied by a timestamp.

Types of Monitoring

- Basic monitoring is automatically enabled for most AWS services, offering metrics at 5-minute intervals for free^s.

- Detailed monitoring, which requires manual activation and incurs extra costs, provides metrics at 1-minute intervals.
- Metric resolution allows for different granularities, with standard resolution offering 1-minute granularity and high resolution allowing 1-second accuracies.

Navigating the CloudWatch Console

- In the CloudWatch console, users can find metrics under different namespaces and filter specific instances.
- Users can adjust time ranges, zoom in on metrics, and change how data is displayed (line chart, stacked area, etc.)^{SS}.
- Alarms can be set to trigger emails or messages based on specific metrics, such as CPU utilization reaching a certain threshold^S.

Detailed Monitoring Activation

- To enable detailed monitoring, it must be done at the service level. For example, in EC2, detailed monitoring can be activated via the management settings for specific instances.
- It's important to note that this option comes with additional costs.

1. What is Amazon CloudWatch primarily used for
2. What are namespaces in Amazon CloudWatch?
3. What are the two types of monitoring available in AWS CloudWatch?
4. What is the difference between standard resolution and high resolution in CloudWatch metrics?
5. What is the difference between basic monitoring and detailed monitoring in Amazon CloudWatch?
6. What does 'resolution' refer to in the context of Amazon CloudWatch metrics?
7. What should you do if you can't find a specific metric in Amazon CloudWatch?
8. How can you narrow down metrics to a specific EC2 instance in CloudWatch?
9. How can you filter metrics for a specific CloudWatch instance?
10. What should you do if you don't see any metrics for your CloudWatch instance?
11. What is the default monitoring resolution in Amazon CloudWatch when using basic monitoring?
12. How can you change the way data is displayed in Amazon CloudWatch?
13. What are some actions you can take with CloudWatch metrics in the AWS console?
14. How can you enable detailed monitoring for an EC2 instance in CloudWatch?
15. What is Amazon CloudWatch used for?
16. How can you access CloudWatch metrics in the AWS console?
17. What is Amazon CloudWatch used for?
18. How can users navigate the CloudWatch console effectively?